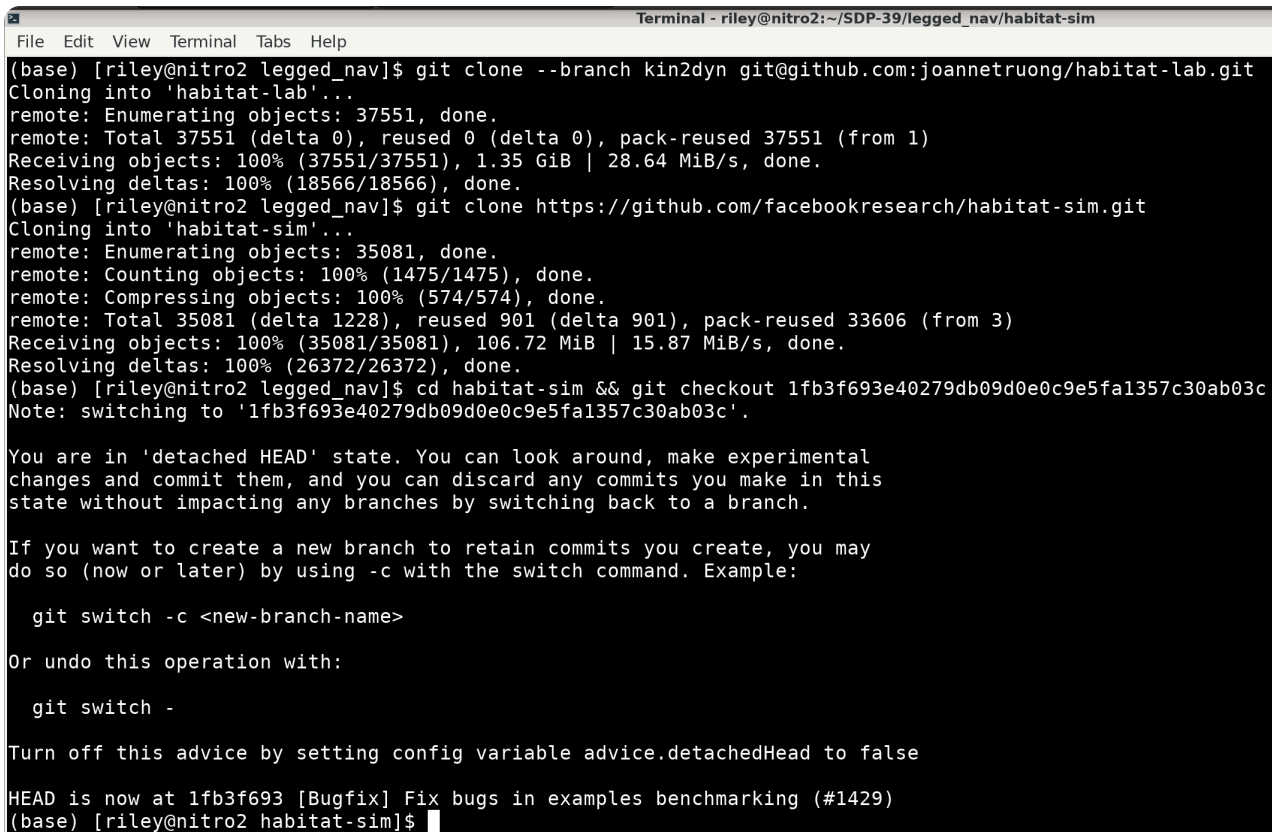


Clone the repos

```
mkdir legged_nav
cd legged_nav
git clone --branch kin2dyn git@github.com:joannetruong/habitat-lab.git
git clone https://github.com/facebookresearch/habitat-sim.git
cd habitat-sim && git checkout 1fb3f693e40279db09d0e0c9e5fa1357c30ab03c
```

A terminal window titled "Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-sim" showing the execution of git commands. The user clones two repositories: 'habitat-lab' and 'habitat-sim'. After cloning 'habitat-sim', they run 'cd habitat-sim && git checkout 1fb3f693e40279db09d0e0c9e5fa1357c30ab03c'. The terminal output shows progress bars for cloning and checkout, and a message indicating the user is in a 'detached HEAD' state. It also provides instructions on how to switch branches and how to turn off the detached head advice.

```
File Edit View Terminal Tabs Help
Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-sim
(base) [riley@nitro2 legged_nav]$ git clone --branch kin2dyn git@github.com:joannetruong/habitat-lab.git
Cloning into 'habitat-lab'...
remote: Enumerating objects: 37551, done.
remote: Total 37551 (delta 0), reused 0 (delta 0), pack-reused 37551 (from 1)
Receiving objects: 100% (37551/37551), 1.35 GiB | 28.64 MiB/s, done.
Resolving deltas: 100% (18566/18566), done.
(base) [riley@nitro2 legged_nav]$ git clone https://github.com/facebookresearch/habitat-sim.git
Cloning into 'habitat-sim'...
remote: Enumerating objects: 35081, done.
remote: Counting objects: 100% (1475/1475), done.
remote: Compressing objects: 100% (574/574), done.
remote: Total 35081 (delta 1228), reused 901 (delta 901), pack-reused 33606 (from 3)
Receiving objects: 100% (35081/35081), 106.72 MiB | 15.87 MiB/s, done.
Resolving deltas: 100% (26372/26372), done.
(base) [riley@nitro2 legged_nav]$ cd habitat-sim && git checkout 1fb3f693e40279db09d0e0c9e5fa1357c30ab03c
Note: switching to '1fb3f693e40279db09d0e0c9e5fa1357c30ab03c'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

    git switch -c <new-branch-name>

Or undo this operation with:

    git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 1fb3f693 [Bugfix] Fix bugs in examples benchmarking (#1429)
(base) [riley@nitro2 habitat-sim]$
```

Create the conda environment

```
conda create -n legged_nav -y python=3.7 cmake=3.14.0
conda activate legged_nav
```

```
Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-sim
File Edit View Terminal Tabs Help

libgomp                pkgs/main/linux-64::libgomp-15.2.0-h4751f2c_7
libnghttp2             pkgs/main/linux-64::libnghttp2-1.52.0-ha637b67_1
libssh2               pkgs/main/linux-64::libssh2-1.10.0-h37d81fd_2
libstdcxx             pkgs/main/linux-64::libstdcxx-15.2.0-h39759b7_7
libstdcxx-ng          pkgs/main/linux-64::libstdcxx-ng-15.2.0-hc03a8fd_7
libxcb                pkgs/main/linux-64::libxcb-1.17.0-h9b100fa_0
libzlib               pkgs/main/linux-64::libzlib-1.3.1-hb25bd0a_0
ncurses                pkgs/main/linux-64::ncurses-6.5-h7934f7d_0
openssl               pkgs/main/linux-64::openssl-1.1.1w-h7f8727e_0
pip                   pkgs/main/linux-64::pip-22.3.1-py37h06a4308_0
pthread-stubs         pkgs/main/linux-64::pthread-stubs-0.3-h0ce48e5_1
python                pkgs/main/linux-64::python-3.7.16-h7a1cb2a_0
readline              pkgs/main/linux-64::readline-8.3-hc2a1206_0
rhash                 pkgs/main/linux-64::rhash-1.4.1-h3c74f83_1
setuptools            pkgs/main/linux-64::setuptools-65.6.3-py37h06a4308_0
sqlite                pkgs/main/linux-64::sqlite-3.51.0-h2a70700_0
tk                    pkgs/main/linux-64::tk-8.6.15-h54e0aa7_0
wheel                 pkgs/main/linux-64::wheel-0.38.4-py37h06a4308_0
xorg-libx11           pkgs/main/linux-64::xorg-libx11-1.8.12-h9b100fa_1
xorg-libxau           pkgs/main/linux-64::xorg-libxau-1.0.12-h9b100fa_0
xorg-libxdmcp         pkgs/main/linux-64::xorg-libxdmcp-1.1.5-h9b100fa_0
xorg-xorgproto        pkgs/main/linux-64::xorg-xorgproto-2024.1-h5eee18b_1
xz                    pkgs/main/linux-64::xz-5.6.4-h5eee18b_1
zlib                  pkgs/main/linux-64::zlib-1.3.1-hb25bd0a_0

Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate ln
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) [riley@nitro2 habitat-sim]$ conda activate ln
(ln) [riley@nitro2 habitat-sim]$
```

I'm using a different environment name as to not interfere with my other cpu installation.

Installing Habitat-Sim

```
pip install -r requirements.txt
python setup.py install --bullet --headless
```

Here I am not using the `--headless` flag.

`pip install -r requirements.txt :`

```
Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-sim
File Edit View Terminal Tabs Help
Using cached numpy-1.21.6-cp37-cp37m-manylinux_2_12_x86_64.manylinux2010_x86_64.whl (15.7 MB)
Collecting numpy-quaternion
Using cached numpy_quaternion-2023.0.4-cp37-cp37m-linux_x86_64.whl
Collecting pillow
Using cached Pillow-9.5.0-cp37-cp37m-manylinux_2_28_x86_64.whl (3.4 MB)
Collecting scipy>=1.3.0
Using cached scipy-1.7.3-cp37-cp37m-manylinux_2_12_x86_64.manylinux2010_x86_64.whl (38.1 MB)
Collecting tqdm
Using cached tqdm-4.67.1-py3-none-any.whl (78 kB)
Collecting importlib-metadata
Using cached importlib_metadata-6.7.0-py3-none-any.whl (22 kB)
Collecting typing-extensions>=3.10.0.2
Using cached typing_extensions-4.7.1-py3-none-any.whl (33 kB)
Collecting gitdb<5,>=4.0.1
Using cached gitdb-4.0.12-py3-none-any.whl (62 kB)
Requirement already satisfied: setuptools in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from imageio-ffmpeg->-r requirements.txt (line 4)) (65.6.3)
Collecting cycler>=0.10
Using cached cycler-0.11.0-py3-none-any.whl (6.4 kB)
Collecting pyparsing>=2.2.1
Using cached pyparsing-3.1.4-py3-none-any.whl (104 kB)
Collecting fonttools>=4.22.0
Using cached fonttools-4.38.0-py3-none-any.whl (965 kB)
Collecting packaging>=20.0
Using cached packaging-24.0-py3-none-any.whl (53 kB)
Collecting python-dateutil>=2.7
Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Collecting kiwisolver>=1.0.1
Using cached kiwisolver-1.4.5-cp37-cp37m-manylinux_2_5_x86_64.manylinux1_x86_64.whl (1.1 MB)
Collecting llvmlite<0.40,>=0.39.0dev0
Using cached llvmlite-0.39.1-cp37-cp37m-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (34.6 MB)
Collecting smmap<6,>=3.0.1
Using cached smmap-5.0.2-py3-none-any.whl (24 kB)
Collecting six>=1.5
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Collecting zipp>=0.5
Using cached zipp-3.15.0-py3-none-any.whl (6.8 kB)
Installing collected packages: zipp, typing-extensions, tqdm, smmap, six, pyparsing, pillow, packaging, numpy, llvmlite, imageio-ffmpeg, fonttools, cycler, scipy, python-dateutil, numpy-quaternion, kiwisolver, importlib-metadata, imageio, gitdb, numba, matplotlib, gitpython, attrs
Successfully installed attrs-24.2.0 cycler-0.11.0 fonttools-4.38.0 gitdb-4.0.12 gitpython-3.1.45 imageio-2.31.2 imageio-ffmpeg-0.5.1 importlib-metadata-6.7.0 kiwisolver-1.4.5 llvmlite-0.39.1 matplotlib-3.5.3 numba-0.56.4 numpy-1.21.6 numpy-quaternion-2023.0.4 packaging-24.0 pillow-9.5.0 pyparsing-3.1.4 python-dateutil-2.9.0.post0 scipy-1.7.3 six-1.17.0 smmap-5.0.2 tqdm-4.67.1 typing-extensions-4.7.1 zipp-3.15.0
(ln) [riley@nitro2 habitat-sim]$
```

`python setup.py install --bullet`

```
Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-sim
File Edit View Terminal Tabs Help
Using /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages
Searching for cycler==0.11.0
Best match: cycler 0.11.0
Adding cycler 0.11.0 to easy-install.pth file

Using /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages
Searching for typing-extensions==4.7.1
Best match: typing-extensions 4.7.1
Adding typing-extensions 4.7.1 to easy-install.pth file

Using /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages
Searching for gitdb==4.0.12
Best match: gitdb 4.0.12
Adding gitdb 4.0.12 to easy-install.pth file

Using /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages
Searching for zipp==3.15.0
Best match: zipp 3.15.0
Adding zipp 3.15.0 to easy-install.pth file

Using /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages
Searching for six==1.17.0
Best match: six 1.17.0
Adding six 1.17.0 to easy-install.pth file

Using /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages
Searching for smmap==5.0.2
Best match: smmap 5.0.2
Adding smmap 5.0.2 to easy-install.pth file

Using /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages
Finished processing dependencies for habitat-sim==0.2.1
Processing ./build/deps/magnum-bindings/src/python
  Preparing metadata (setup.py) ... done
Building wheels for collected packages: magnum
  Building wheel for magnum (setup.py) ... done
  Created wheel for magnum: filename=magnum-0.0.0-cp37-cp37m-linux_x86_64.whl size=45639807 sha256=1bf0a3e5ad85b60357c3aef29e2fddc0db48e30bb8d998f337bcbd3bdd89e81c
  Stored in directory: /tmp/pip-ephem-wheel-cache-o2jmq6mb/wheels/f6/cd/82/a94984b2fd7ff8820b8c89e21c14ab5f9b514e11c04770fed5
Successfully built magnum
Installing collected packages: magnum
Successfully installed magnum-0.0.0
(ln) [riley@nitro2 habitat-sim]$
```

This command seems to have finished without a problem.

Installing Habitat-Lab

```
cd ../habitat-lab
pip install typing-extensions~=3.7.4 google-auth==1.6.3 simplejson braceexpand
conda install -y pytorch torchvision torchaudio cudatoolkit=11.3 -c pytorch
pip install -r requirements.txt
python setup.py develop --all
```

```
Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-lab
File Edit View Terminal Tabs Help
(ln) [riley@nitro2 habitat-sim]$ cd ../habitat-lab/
(ln) [riley@nitro2 habitat-lab]$ pip install typing-extensions~=3.7.4 google-auth==1.6.3 simplejson braceexpand pybullet cython pkgconfig squaternion
Collecting typing-extensions~=3.7.4
  Using cached typing_extensions-3.7.4.3-py3-none-any.whl (22 kB)
Collecting google-auth==1.6.3
  Using cached google_auth-1.6.3-py2.py3-none-any.whl (73 kB)
Collecting simplejson
  Using cached simplejson-3.20.2-cp37-cp37m-manylinux_2_5_x86_64.manylinux1_x86_64.manylinux2_17_x86_64.manylinux2014_x86_64.whl (127 kB)
Collecting braceexpand
  Using cached braceexpand-0.1.7-py2.py3-none-any.whl (5.9 kB)
Collecting pybullet
  Using cached pybullet-3.2.7-cp37-cp37m-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (103.2 MB)
Collecting cython
  Using cached Cython-3.0.12-cp37-cp37m-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.6 MB)
Collecting pkgconfig
  Using cached pkgconfig-1.5.5-py3-none-any.whl (6.7 kB)
Collecting squaternion
  Using cached squaternion-0.3.2-py3-none-any.whl (7.3 kB)
Collecting pyasn1-modules>=0.2.1
  Using cached pyasn1_modules-0.3.0-py2.py3-none-any.whl (181 kB)
Requirement already satisfied: six>=1.9.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from google-auth==1.6.3) (1.17.0)
Collecting cachetools>=2.0.0
  Using cached cachetools-5.5.2-py3-none-any.whl (10 kB)
Collecting rsa>=3.1.4
  Using cached rsa-4.9.1-py3-none-any.whl (34 kB)
Requirement already satisfied: importlib-metadata in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from squaternion) (6.7.0)
Collecting pyasn1<0.6.0,>=0.4.6
  Using cached pyasn1-0.5.1-py2.py3-none-any.whl (84 kB)
Requirement already satisfied: zipp>=0.5 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from importlib-metadata->squaternion) (3.15.0)
Installing collected packages: typing-extensions, pybullet, braceexpand, simplejson, pyasn1, pkgconfig, cython, cachetools, squaternion, rsa, pyasn1-modules, google-auth
  Attempting uninstall: typing-extensions
    Found existing installation: typing_extensions 4.7.1
    Uninstalling typing_extensions-4.7.1:
      Successfully uninstalled typing_extensions-4.7.1
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
  gitpython 3.1.45 requires typing-extensions>=3.10.0.2; python_version < "3.10", but you have typing-extensions 3.7.4.3 which is incompatible.
Successfully installed braceexpand-0.1.7 cachetools-5.5.2 cython-3.0.12 google-auth-1.6.3 pkgconfig-1.5.5 pyasn1-0.5.1 pyasn1-modules-0.3.0 pybullet-3.2.7 rsa-4.9.1 simplejson-3.20.2 squaternion-0.3.2 typing-extensions-3.7.4.3
(ln) [riley@nitro2 habitat-lab]$
```

`typing_extensions` errors during installation.

Using `pip install torch==1.13.1+cu117 torchvision==0.14.1+cu117 torchaudio==0.13.1 --extra-index-url https://download.pytorch.org/whl/cu117` instead of the given conda command:

```

Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-lab
File Edit View Terminal Tabs Help
(ln) [riley@nitro2 habitat-lab]$ pip install torch==1.13.1+cu117 torchvision==0.14.1+cu117 torchaudio==0.13.1 --extra-index-url https://download.pytorch.org/whl/cu117
Looking in indexes: https://pypi.org/simple, https://download.pytorch.org/whl/cu117
Collecting torch==1.13.1+cu117
  Using cached https://download.pytorch.org/whl/cu117/torch-1.13.1%2Bcu117-cp37-cp37m-linux_x86_64.whl (1801.9 MB)
Collecting torchvision==0.14.1+cu117
  Using cached https://download.pytorch.org/whl/cu117/torchvision-0.14.1%2Bcu117-cp37-cp37m-linux_x86_64.whl (24.3 MB)
Collecting torchaudio==0.13.1
  Using cached https://download.pytorch.org/whl/cu117/torchaudio-0.13.1%2Bcu117-cp37-cp37m-linux_x86_64.whl (4.2 MB)
Requirement already satisfied: typing-extensions in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from torch==1.13.1+cu117) (3.7.4.3)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from torchvision==0.14.1+cu117) (9.5.0)
Requirement already satisfied: numpy in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from torchvision==0.14.1+cu117) (1.21.6)
Collecting requests
  Using cached requests-2.31.0-py3-none-any.whl (62 kB)
Requirement already satisfied: certifi>=2017.4.17 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from requests->torchvision==0.14.1+cu117) (2022.12.7)
Collecting urllib3<3,>=1.21.1
  Using cached urllib3-2.0.7-py3-none-any.whl (124 kB)
Collecting charset-normalizer<4,>=2
  Using cached charset-normalizer-3.4.4-py3-none-any.whl (53 kB)
Collecting idna<4,>=2.5
  Using cached idna-3.10-py3-none-any.whl (70 kB)
Installing collected packages: urllib3, torch, idna, charset-normalizer, torchaudio, requests, torchvision
Successfully installed charset-normalizer-3.4.4 idna-3.10 requests-2.31.0 torch-1.13.1+cu117 torchaudio-0.13.1+cu117 torchvision-0.14.1+cu117 urllib3-2.0.7
(ln) [riley@nitro2 habitat-lab]$

```

```

Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-lab
File Edit View Terminal Tabs Help
Requirement already satisfied: numpy>=1.16.1 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from -r requirements.txt (line 2)) (1.21.6)
Collecting yacs>=0.1.5
  Using cached yacs-0.1.8-py3-none-any.whl (14 kB)
Requirement already satisfied: numpy-quaternion>=2019.3.18.14.33.20 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from -r requirements.txt (line 4)) (2023.0.4)
Requirement already satisfied: attrs>=19.1.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from -r requirements.txt (line 5)) (24.2.0)
Collecting opencv-python>=3.3.0
  Using cached opencv_python-4.12.0.88-cp37-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (67.0 MB)
Collecting pickle5
  Using cached pickle5-0.0.12-cp37-cp37m-manylinux_2_5_x86_64.manylinux1_x86_64.whl (256 kB)
Requirement already satisfied: imageio>=2.2.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from -r requirements.txt (line 9)) (2.31.2)
Requirement already satisfied: imageio-ffmpeg>=0.2.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from -r requirements.txt (line 10)) (0.5.1)
Requirement already satisfied: scipy>=1.0.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from -r requirements.txt (line 11)) (1.7.3)
Requirement already satisfied: tqdm>=4.0.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from -r requirements.txt (line 12)) (4.67.1)
Requirement already satisfied: numba>=0.44.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from -r requirements.txt (line 13)) (0.56.4)
Collecting gym-notices>=0.0.4
  Using cached gym_notices-0.1.0-py3-none-any.whl (3.3 kB)
Requirement already satisfied: importlib-metadata>=4.8.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from gym>=0.17.3->-r requirements.txt (line 1)) (6.7.0)
Collecting cloudpickle>=1.2.0
  Using cached cloudpickle-2.2.1-py3-none-any.whl (25 kB)
Collecting PyYAML
  Using cached PyYAML-6.0.1-cp37-cp37m-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (670 kB)
Requirement already satisfied: pillow>=8.3.2 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from imageio>=2.2.0->-r requirements.txt (line 9)) (9.5.0)
Requirement already satisfied: setuptools in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from imageio-ffmpeg>=0.2.0->-r requirements.txt (line 10)) (65.6.3)
Requirement already satisfied: llvmlite<0.40,>=0.39.0dev0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from numba>=0.44.0->-r requirements.txt (line 13)) (0.39.1)
Requirement already satisfied: zipp>=0.5 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from importlib-metadata>=4.8.0->gym>=0.17.3->-r requirements.txt (line 1)) (3.15.0)
Requirement already satisfied: typing-extensions>=3.6.4 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from importlib-metadata>=4.8.0->gym>=0.17.3->-r requirements.txt (line 1)) (3.7.4.3)
Installing collected packages: gym-notices, PyYAML, pickle5, opencv-python, cloudpickle, yacs, gym
Successfully installed PyYAML-6.0.1 cloudpickle-2.2.1 gym-0.26.2 gym-notices-0.1.0 opencv-python-4.12.0.88 pickle5-0.0.12 yacs-0.1.8
(ln) [riley@nitro2 habitat-lab]$

```

Using `python setup.py install --all` instead of develop:

```
Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-lab
File Edit View Terminal Tabs Help
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat/utils/visualizations/fog_of_war.py to fog_of_war.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat/utils/visualizations/map5.py to maps.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat/utils/visualizations/utlis.py to utlis.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/_init_.py to _init_.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/run.py to run.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/agents/_init_.py to _init_.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/agents/ppo_agents.py to ppo_agents.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/agents/simple_agents.py to simple_agents.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/agents/slam_agents.py to slam_agents.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/config/_init_.py to _init_.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/config/default.py to default.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/_init_.py to _init_.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ddppo/_init_.py to _init_.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ddppo/ddp_utils.py to ddp_utils.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ddppo/algo/_init_.py to _init_.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ddppo/algo/ddppo.py to ddppo.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ddppo/policy/_init_.py to _init_.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ddppo/policy/resnet.py to resnet.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ddppo/policy/resnet_policy.py to resnet_policy.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ddppo/policy/running_mean_and_var.py to running_mean_and_var.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/models/_init_.py to _init_.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/models/rnn_state_encoder.py to rnn_state_encoder.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/models/simple_cnn.py to simple_cnn.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/models/z_model.py to z_model.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ppo/_init_.py to _init_.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ppo/policy.py to policy.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ppo/ppo.py to ppo.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/rl/ppo/ppo_trainer.py to ppo_trainer.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/utlis/_init_.py to _init_.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/utlis/common.py to common.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/utlis/env_utlis.py to env_utlis.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/utlis/visualizations/_init_.py to _init_.cpython-37.pyc
byte-compiling /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat_baselines/utlis/visualizations/utlis.py to utlis.cpython-37.pyc
running install egg info
Copying habitat.egg-info to /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/habitat-0.1.7-py3.7.egg-info
running install scripts
(ln) [riley@nitro2 habitat-lab]$
```

Adjusting YAML files.

In `pointnav_quadrupe`:

- POSITION / ORIENTATION changes as described.
- ROBOT_URF -> `/shared/habitat_datasets/URDF_demo_assets/spot/urdf/spot.urdf`
- DATA_PATH -> `/shared/habitat_datasets/datasets/pointnav_hm3d_gibson/pointnav_spot_0.3/{split}/{split}.json.gz`

In `ddppo_pointnav_quadrupe`:

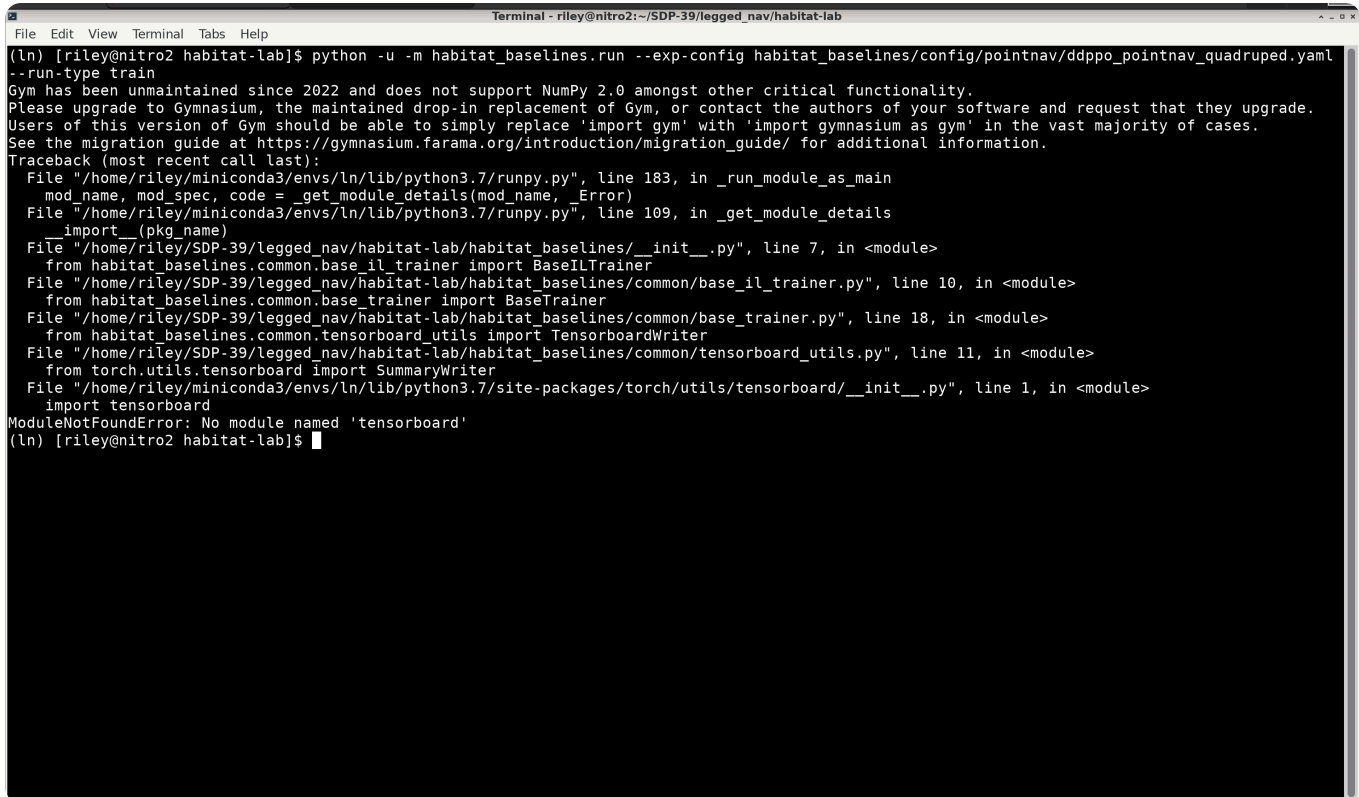
```
TENSORBOARD_DIR: './output/tb'
VIDEO_DIR: './output/video_dir'
EVAL_CKPT_PATH_DIR: './output/checkpoints'
CHECKPOINT_FOLDER: './output/checkpoints'
TXT_DIR: './output/txts'
```

Training

I use the following command:

```
python -u -m habitat_baselines.run --exp-config habitat_baselines/config/pointnav/ddppo_pointnav_quadruped.yaml
```

Errors

A terminal window titled "Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-lab" shows the execution of a Python command. The command is `python -u -m habitat\_baselines.run --exp-config habitat\_baselines/config/pointnav/ddppo\_pointnav\_quadruped.yaml --run-type train`. The output shows a warning about Gym being unmaintained and a message about upgrading to Gymnasium. A traceback follows, showing the error originates from the `\_\_import\_\_` function in the `\_\_init\_\_.py` file of the `habitat\_baselines` module. The error message is `ModuleNotFoundError: No module named 'tensorboard'`. The terminal prompt is `(ln) [riley@nitro2 habitat-lab]\$`.

```
File Edit View Terminal Tabs Help
(ln) [riley@nitro2 habitat-lab]$ python -u -m habitat_baselines.run --exp-config habitat_baselines/config/pointnav/ddppo_pointnav_quadruped.yaml
--run-type train
Gym has been unmaintained since 2022 and does not support NumPy 2.0 amongst other critical functionality.
Please upgrade to Gymnasium, the maintained drop-in replacement of Gym, or contact the authors of your software and request that they upgrade.
Users of this version of Gym should be able to simply replace 'import gym' with 'import gymnasium as gym' in the vast majority of cases.
See the migration guide at https://gymnasium.farama.org/introduction/migration_guide/ for additional information.
Traceback (most recent call last):
  File "/home/riley/miniconda3/envs/ln/lib/python3.7/runpy.py", line 183, in _run_module_as_main
    mod_name, mod_spec, code = _get_module_details(mod_name, _Error)
  File "/home/riley/miniconda3/envs/ln/lib/python3.7/runpy.py", line 109, in _get_module_details
    __import__(pkg_name)
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/__init__.py", line 7, in <module>
    from habitat_baselines.common.base_il_trainer import BaseILTrainer
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/common/base_il_trainer.py", line 10, in <module>
    from habitat_baselines.common.base_trainer import BaseTrainer
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/common/base_trainer.py", line 18, in <module>
    from habitat_baselines.common.tensorboard_utils import TensorboardWriter
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/common/tensorboard_utils.py", line 11, in <module>
    from torch.utils.tensorboard import SummaryWriter
  File "/home/riley/miniconda3/envs/ln/lib/python3.7/site-packages/torch/utils/tensorboard/__init__.py", line 1, in <module>
    import tensorboard
ModuleNotFoundError: No module named 'tensorboard'
(ln) [riley@nitro2 habitat-lab]$
```

Tensorboard issue, fixed using `pip install tensorboard`.

```

Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-lab
File Edit View Terminal Tabs Help
Collecting grpcio>=1.24.3
  Using cached grpcio-1.62.3-cp37-cp37m-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (5.6 MB)
Collecting protobuf<4,>=3.9.2
  Using cached protobuf-3.20.3-cp37-cp37m-manylinux_2_5_x86_64.manylinux1_x86_64.whl (1.0 MB)
Requirement already satisfied: numpy>=1.12.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from tensorboard) (1.21.6)
Requirement already satisfied: requests<3,>=2.21.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from tensorboard) (2.31.0)
Collecting tensorboard-data-server<0.7.0,>=0.6.0
  Using cached tensorboard_data_server-0.6.1-py3-none-manylinux2010_x86_64.whl (4.9 MB)
Requirement already satisfied: cachetools>=2.0.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from google-auth<3,>=1.6.3->tensorboard) (5.5.2)
Requirement already satisfied: rsa>=3.1.4 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from google-auth<3,>=1.6.3->tensorboard) (4.9.1)
Requirement already satisfied: pyasn1-modules>=0.2.1 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from google-auth<3,>=1.6.3->tensorboard) (0.3.0)
Requirement already satisfied: six>=1.9.0 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from google-auth<3,>=1.6.3->tensorboard) (1.17.0)
Collecting requests-oauthlib>=0.7.0
  Using cached requests_oauthlib-2.0.0-py2.py3-none-any.whl (24 kB)
Requirement already satisfied: importlib-metadata>=4 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from markdown>=2.6.8->tensorboard) (6.7.0)
Requirement already satisfied: urllib3<3,>=1.21.1 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from requests<3,>=2.21.0->tensorboard) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from requests<3,>=2.21.0->tensorboard) (2022.12.7)
Requirement already satisfied: idna<4,>=2.5 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from requests<3,>=2.21.0->tensorboard) (3.10)
Requirement already satisfied: charset-normalizer<4,>=2 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from requests<3,>=2.21.0->tensorboard) (3.4.4)
Collecting MarkupSafe>=2.1.1
  Using cached MarkupSafe-2.1.5-cp37-cp37m-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (25 kB)
Requirement already satisfied: typing-extensions>=3.6.4 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard) (3.7.4.3)
Requirement already satisfied: zipp>=0.5 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard) (3.15.0)
Requirement already satisfied: pyasn1<0.6.0,>=0.4.6 in /home/riley/miniconda3/envs/ln/lib/python3.7/site-packages (from pyasn1-modules>=0.2.1->google-auth<3,>=1.6.3->tensorboard) (0.5.1)
Collecting oauthlib>=3.0.0
  Using cached oauthlib-3.2.2-py3-none-any.whl (151 kB)
Installing collected packages: tensorboard-plugin-wit, tensorboard-data-server, protobuf, oauthlib, MarkupSafe, grpcio, absl-py, werkzeug, requests-oauthlib, markdown, google-auth-oauthlib, tensorboard
Successfully installed MarkupSafe-2.1.5 absl-py-2.1.0 google-auth-oauthlib-0.4.6 grpcio-1.62.3 markdown-3.4.4 oauthlib-3.2.2 protobuf-3.20.3 requests-oauthlib-2.0.0 tensorboard-2.11.2 tensorboard-data-server-0.6.1 tensorboard-plugin-wit-1.8.1 werkzeug-2.2.3
(ln) [riley@nitro2 habitat-lab]$

```

3

```

Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-lab
File Edit View Terminal Tabs Help
(ln) [riley@nitro2 habitat-lab]$ python -u -m habitat_baselines.run --exp-config habitat_baselines/config/pointnav/ddppo_pointnav_quadruped.yaml --run-type train
Gym has been unmaintained since 2022 and does not support NumPy 2.0 amongst other critical functionality.
Please upgrade to Gymnasium, the maintained drop-in replacement of Gym, or contact the authors of your software and request that they upgrade.
Users of this version of Gym should be able to simply replace 'import gym' with 'import gymnasium as gym' in the vast majority of cases.
See the migration guide at https://gymnasium.farama.org/introduction/migration_guide/ for additional information.
Traceback (most recent call last):
  File "/home/riley/miniconda3/envs/ln/lib/python3.7/runpy.py", line 183, in _run_module_as_main
    mod_name, mod_spec, code = _get_module_details(mod_name, _Error)
  File "/home/riley/miniconda3/envs/ln/lib/python3.7/runpy.py", line 109, in _get_module_details
    __import__(pkg_name)
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/_init__.py", line 7, in <module>
    from habitat_baselines.common.base_il_trainer import BaseILTrainer
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/common/base_il_trainer.py", line 10, in <module>
    from habitat_baselines.common.base_trainer import BaseTrainer
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/common/base_trainer.py", line 19, in <module>
    from habitat_baselines.rl.ddppo.ddp_utils import SAVE_STATE, is_slurm_batch_job
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/rl/ddppo/ddp_utils.py", line 9, in <module>
    import ifcfg
ModuleNotFoundError: No module named 'ifcfg'
(ln) [riley@nitro2 habitat-lab]$

```

`ifcfg` package is not found. Fixed using `pip install ifcfg`. This does throw errors when installing though.

```

ModuleNotFoundError: No module named 'ifcfg'
(ln) [riley@nitro2 habitat-lab]$ pip install ifcfg
Collecting ifcfg
  Using cached ifcfg-0.24-py2.py3-none-any.whl (10 kB)
Installing collected packages: ifcfg
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
habitat 0.1.7 requires lmbd>=0.98, which is not installed.
habitat 0.1.7 requires moviepy>=1.0.1, which is not installed.
habitat 0.1.7 requires tb-nightly, which is not installed.
habitat 0.1.7 requires tensorflow==1.13.1, which is not installed.
habitat 0.1.7 requires webdataset==0.1.40, which is not installed.
Successfully installed ifcfg-0.24
(ln) [riley@nitro2 habitat-lab]$

```

3

Finally we have all the packages installed, however it still does not run because it can't seem to see the dataset.

```

Terminal - riley@nitro2:~/SDP-39/legged_nav/habitat-lab
File Edit View Terminal Tabs Help
(riley@nitro2 habitat-lab)$ python -u -m habitat_baselines.run --exp-config habitat_baselines/config/pointnav/ddppo_pointnav_quadrupe
--run-type train
Gym has been unmaintained since 2022 and does not support NumPy 2.0 amongst other critical functionality.
Please upgrade to Gymnasium, the maintained drop-in replacement of Gym, or contact the authors of your software and request that they upgrade.
Users of this version of Gym should be able to simply replace 'import gym' with 'import gymnasium as gym' in the vast majority of cases.
See the migration guide at https://gymnasium.farama.org/introduction/migration_guide/ for additional information.
2025-11-18 11:50:05.246 Initializing dataset PointNav-v1
DATA PATH: /shared/habitat_datasets/datasets/pointnav_hm3d_gibson/pointnav_spot_0.3/train/train.json.gz
Traceback (most recent call last):
  File "/home/riley/miniconda3/envs/ln/lib/python3.7/runpy.py", line 193, in _run_module_as_main
    "__main__", mod_spec)
  File "/home/riley/miniconda3/envs/ln/lib/python3.7/runpy.py", line 85, in _run_code
    exec(code, run_globals)
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/run.py", line 94, in <module>
    main()
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/run.py", line 40, in main
    run_exp(**vars(args))
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/run.py", line 90, in run_exp
    execute_exp(config, run_type)
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/run.py", line 60, in execute_exp
    trainer.train()
  File "/home/riley/miniconda3/envs/ln/lib/python3.7/contextlib.py", line 74, in inner
    return func(*args, **kwargs)
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/rl/ppo/ppo_trainer.py", line 712, in train
    self._init_train()
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/rl/ppo/ppo_trainer.py", line 259, in _init_train
    self._init_envs()
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/rl/ppo/ppo_trainer.py", line 213, in _init_envs
    workers_ignore_signals=is_slurm_batch_job(),
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat_baselines/utils/env_utils.py", line 59, in construct_envs
    scenes = dataset.get_scenes_to_load(config.TASK_CONFIG.DATASET)
  File "/home/riley/SDP-39/legged_nav/habitat-lab/habitat/datasets/pointnav/pointnav_dataset.py", line 44, in get_scenes_to_load
    assert cls.check_config_paths_exist(config)
AssertionError
(riley@nitro2 habitat-lab)$

```